

## THERMACART · REQUIREMENTS

# ThermaCart requirements

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## DOCUMENT CONTROL

|             |                                         |         |                |
|-------------|-----------------------------------------|---------|----------------|
| Document ID | TCT-REQ-001                             | Version | 0.4            |
| Status      | <b>Draft</b>                            | Date    | 2026-09-25     |
| Author      | Amish Chadha                            | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept           |         |                |
| Repository  | github.com/BoujeeEnjinia1701/thermacart |         |                |
| Source      | docs/03-requirements.md @ 390f06f       |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                   |
|---------|------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                                                                              |
| 0.3     | 2026-09-25 | Amish Chadha | TRL 3 status from TCT-CAL-001; R12 oven temperature corrected to the RT 70 HC limit; R16 budget scope per TCT-DDR-001 D6 |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates                                       |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                                                                         |

# ThermaCart requirements

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These are the requirements for the TC-L cartridge and its adapter frame. The targets were proposed at TRL 2, and the design choices behind them were decided by Amish on 2026-09-25 (TCT-DDR-001 and TCT-DDR-002). Under TCT-DDR-002 the C25 band of R1 is widened to 22 to 27 °C (O9), R5 is relaxed to 38 % for the stock-tube build with 50 % kept as the target for a later extrusion (O4), and R14 carries a labelled handling rule for H70 cartridges taken from an oven (O7). R10 keeps its 8 h target (O5). The status column comes from TCT-CAL-001 v0.2, which uses the decided dark finish (O8). Two requirements are **not met** on paper (R10 and R16), one is **at risk** (R12), three can only be settled by tests (R6, R7 and R17) and twelve are met.

Table 1. Requirements

| ID  | Requirement                                          | Target                                                                                                                                                   | Verification (TRL 3 or later)                                     | Status at TRL 3 (TCT-CAL-001)                                                                                                                                                                                                                                                  |
|-----|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Offer a small set of standard temperature grades     | At least three: C5 (melt 4 to 6 °C, cold chain), C25 (22 to 27 °C, comfort; widened from 24 to 27 °C by TCT-DDR-002, O9), H70 (68 to 72 °C, hot holding) | Material datasheets; later DSC test of each fill batch            | Met: RT 5 HC 5 to 6 °C, RT 25 HC 22 to 26 °C, RT 70 HC 69 to 71 °C                                                                                                                                                                                                             |
| R2  | Common envelope that fits kitchen and cooler formats | Fits a GN 1/3 slot: at most 325 x 176 mm footprint and 65 mm high, handle included                                                                       | Model and drawing                                                 | Met: 323.4 x 152.4 x 63.5 mm with the bail stowed (TCT-DWG-001)                                                                                                                                                                                                                |
| R3  | Enough stored energy per cartridge                   | 60 Wh (216 kJ) or more of usable storage within $\pm 3$ K of the melt point                                                                              | Calculation per grade                                             | Met: C5 65.4 Wh, C25 60.5 Wh (thin margin), H70 72.9 Wh                                                                                                                                                                                                                        |
| R4  | Carry with one hand                                  | Filled mass 3.0 kg or less                                                                                                                               | Mass estimate, then weighing                                      | Met, thin margin: 2.90 to 2.98 kg                                                                                                                                                                                                                                              |
| R5  | Efficient use of mass                                | PCM at least 38 % of filled mass for the stock-tube build; 50 % remains the target for a later custom extrusion (relaxed by TCT-DDR-002, O4)             | Mass estimate                                                     | Met, thin margin: 38.1 to 39.7 %                                                                                                                                                                                                                                               |
| R6  | Sealed for life                                      | No leak through 1,000 thermal cycles and a 1 m drop onto concrete in any state                                                                           | Seal design review at TRL 3; cycle and drop tests belong to TRL 4 | Not verifiable at TRL 3; an end-on drop of a molten cartridge could see a surge of up to 5 bar                                                                                                                                                                                 |
| R7  | Long life                                            | 1,000 cycles or more with less than 10 % loss of latent capacity                                                                                         | Literature and supplier data at TRL 3                             | Not verifiable at TRL 3; plausible for paraffins                                                                                                                                                                                                                               |
| R8  | No pressure hazard                                   | Internal gauge pressure 0.5 bar or less at the highest charge temperature; ullage of at least 10 % of the inner volume at that temperature               | Calculation                                                       | Met: 0.00 bar at the fill temperature, +0.29 bar at 20 K above the limit; wall stress 56 MPa under the cold vacuum                                                                                                                                                             |
| R9  | Wrong grade cannot be fitted                         | Each grade has its own nose key; a frame for one grade rejects the others                                                                                | Design review of key geometry                                     | Met by design: every pairing checked                                                                                                                                                                                                                                           |
| R10 | Cold-chain hold time                                 | Two C5 cartridges keep a 25 L cooler (heat leak about 0.5 W/K, estimate) at 2 to 8 °C for 8 h or more at 32 °C ambient                                   | Heat balance calculation                                          | <b>Not met:</b> 2.4 h with the decided dark finish (1.9 h bare); the stored energy would last 9.7 h, but air-side heat transfer limits the hold. Produce laid directly on the cartridges (TCT-DDR-002, O5) is not credited and waits for a TRL 4 measurement, which is on hold |
| R11 | Hot-holding time                                     | One H70 cartridge keeps food in an insulated GN carrier (about 0.25 W/K,                                                                                 |                                                                   |                                                                                                                                                                                                                                                                                |

| ID  | Requirement                         | Target                                                                                                                                                                                                                                                            | Verification (TRL 3 or later)                    | Status at TRL 3 (TCT-CAL-001)                                                                                                                                                                                                                |
|-----|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                     | estimate) at 63 °C or more for 4 h or more at 25 °C ambient                                                                                                                                                                                                       | Heat balance calculation                         | Met: 13.6 h with the dark finish (7.2 h bare)                                                                                                                                                                                                |
| R12 | Recharge with common equipment      | Full recharge in 8 h or less: C5 and C25 in a domestic freezer or refrigerator, H70 in an oven set to 85 °C (never above 90 °C) or on a 60 W heating pad                                                                                                          | Calculation (Stefan problem plus fin convection) | <b>At risk:</b> with the dark finish, C5 freezer 4.2 h, C25 refrigerator 4.6 h, H70 oven 5.3 h; H70 pad 2.7 to 10.3 h                                                                                                                        |
| R13 | State of charge visible             | Charged or melted state readable at a glance without tools                                                                                                                                                                                                        | Design review                                    | Met by design (sight window on a PCM vial)                                                                                                                                                                                                   |
| R14 | Safe to handle when charged         | Handle touch temperature below the ISO 13732-1 burn threshold for 10 s contact on every grade; for an H70 cartridge taken from an oven, the label states a handling rule (oven gloves, or wait 12 min) and pad charging is the preferred method (TCT-DDR-002, O7) | Calculation of handle temperature; label review  | Met with the handling rule: bail on a thermal break settles at 42.5 °C on pad charging; after an oven it is at 85 °C and falls below 55 °C in about 12 min, which the label rule covers                                                      |
| R15 | Materials compatible with each fill | No corrosion, swelling or permeation: aluminium shell with FKM or NBR seals for paraffins; coated shell or polymer liner for salt hydrates                                                                                                                        | Compatibility review against supplier data       | Met by design for paraffin grades (epoxy rated 120 °C or more); open for salt hydrates                                                                                                                                                       |
| R16 | Low cost and garage-buildable       | \$50 or less in parts per cartridge and \$20 or less per frame; stock sections, drill press and taps only. The \$80 prototype budget covers one C5 cartridge and one frame (TCT-DDR-001 D6)                                                                       | Priced BOM                                       | <b>Not met:</b> \$76 per cartridge including the dark finish; frame \$16 met; one cartridge and one frame \$92 against \$80 (indicative prices). Supplier quotes come first and the budget is not raised until they are in (TCT-DDR-002, O6) |
| R17 | Open and documented                 | Envelope, key geometry, grade table and fill recipe published under CERN-OHL-S-2.0                                                                                                                                                                                | Repository review                                | Not verifiable at TRL 3; envelope, key and grade tables are published, the fill recipe waits for TRL 4                                                                                                                                       |
| R18 | Recoverable at end of life          | Fill can be drained or melted out through the fill port and the shell reused; fill named on the label                                                                                                                                                             | Design review                                    | Met by design                                                                                                                                                                                                                                |

## ■ Assumptions

- PCM data from the Rubitherm datasheets for RT 5 HC, RT 25 HC and RT 70 HC, taken at the low side of their  $\pm 7.5$  % tolerance; usable storage is the latent heat plus sensible heat within  $\pm 3$  K of the melt point (TCT-CAL-001, Table 1).
- Fill of 1.10 to 1.18 kg: 90 % of a 1.655 L inner volume, filled as liquid at each grade's maximum operating temperature (45, 65 and 90 °C).

- Hot-holding threshold of 63 °C follows common food-safety practice (the US FDA Food Code uses 57 °C, 135 °F); the lower value would lengthen the hold time.
- Cooler and carrier heat leaks are estimates for typical polyurethane and EPP boxes and must be replaced with measured values from a partner's real boxes.
- The burn thresholds used for R14 (about 55 °C for bare metal and 70 °C for plastics, 10 s contact) are commonly quoted from ISO 13732-1 and have not been checked against the standard.